

Class 14: DESeq2 analysis mini project

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Background

Here we work through a complete RNASeq analysis project. The input data comes from a knock-down experiment of a HOX gene.

Data Import

Reading the `counts` and `metadata()` CSV files

```
counts <- read.csv(file = "GSE37704_featurecounts.csv", row.names = 1)
metadata <- read.csv(file = "GSE37704_metadata.csv")
head(counts)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

```
head(metadata)
```

	id	condition
1	SRR493366	control_sirna
2	SRR493367	control_sirna
3	SRR493368	control_sirna
4	SRR493369	hoxa1_kd
5	SRR493370	hoxa1_kd
6	SRR493371	hoxa1_kd

```
nrow(metadata)
```

```
[1] 6
```

```
ncol(counts)
```

```
[1] 7
```

Check on data structure

Some book-keeping is required as there looks to be a mis-match between metadata rows and counts columns.

Looks like we need to get rid of the first “length” column of our `counts` object

```
cleancounts <- counts[,-1]
```

```
all(colnames(cleancounts) == metadata$id)
```

```
[1] TRUE
```

Remove zero count genes

There are lots of genes with zero counts. We can remove these from further analysis.

```
to.keep.ids <- rowSums(cleancounts) > 0  
nonzero_counts <- cleancounts[to.keep.ids,]
```

DESeq analysis

load package

```
library(DESeq2)
```

setup DESeq object

```
dds <- DESeqDataSetFromMatrix(countData = nonzero_counts,  
                              colData = metadata,  
                              design = ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

Run DESeq

```
dds <- DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

get results

```
res <- results(dds)
head(res)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 6 rows and 6 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01
	padj				
	<numeric>				
ENSG00000279457	6.86555e-01				
ENSG00000187634	5.15718e-03				
ENSG00000188976	1.76549e-35				
ENSG00000187961	1.13413e-07				
ENSG00000187583	9.19031e-01				
ENSG00000187642	4.03379e-01				

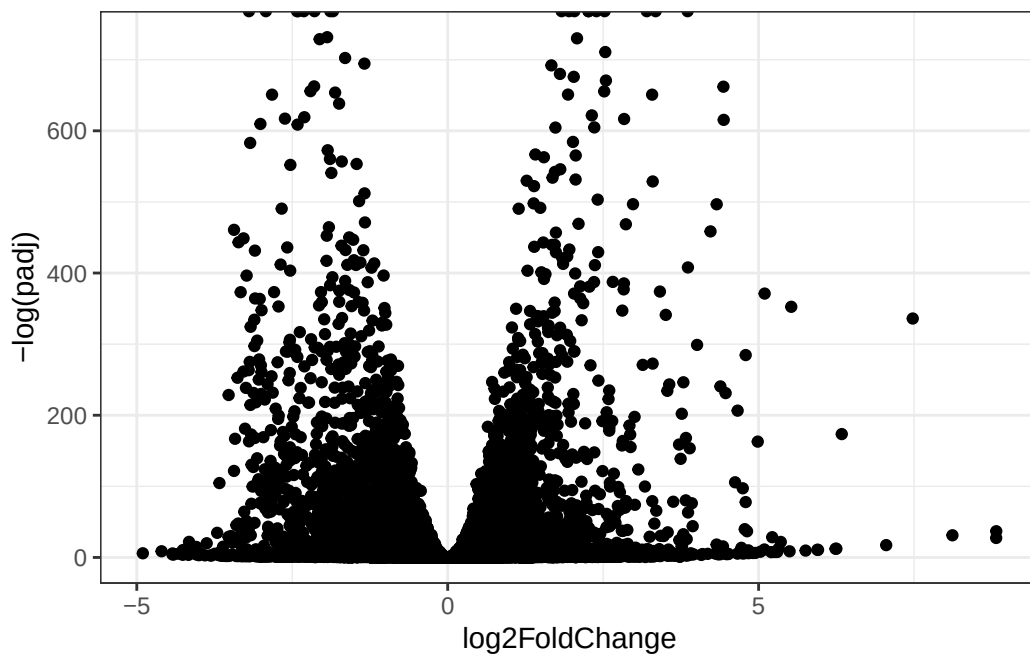
Data Visualization

Volcano plot

```
library(ggplot2)
```

```
ggplot(res)+  
  aes(log2FoldChange,  
      -log(padj))+  
  geom_point()+  
  theme_bw()
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).

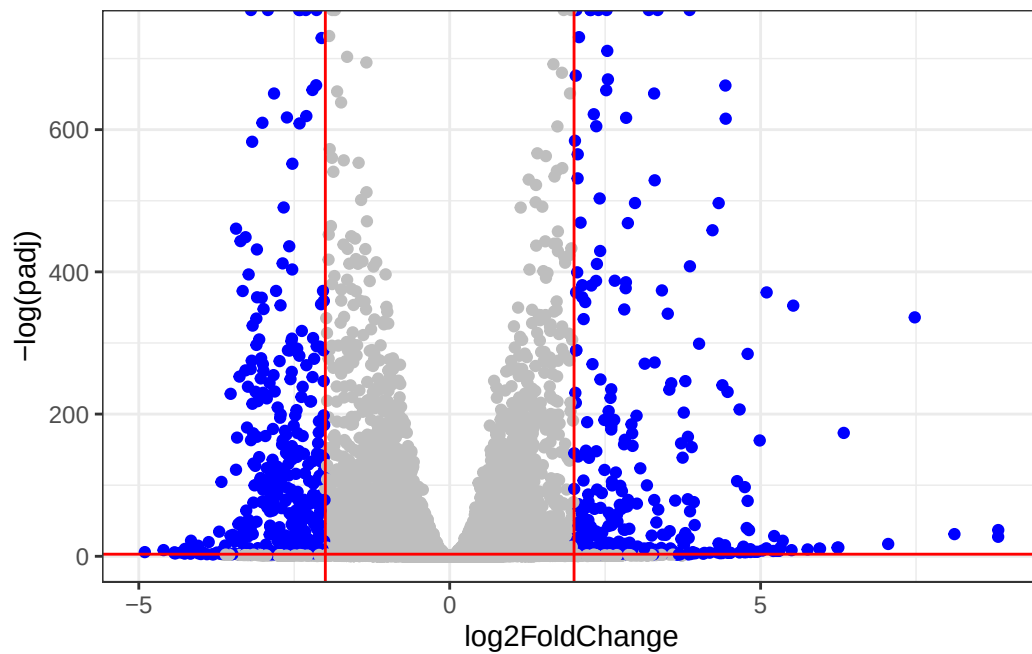


Add threshold lines for fold-change and P-value and color our subset of genes that make these threshold cut-offs in the plot.

```
mycols <- rep("grey",nrow(res))  
mycols[abs(res$log2FoldChange)>2] <- "blue"  
mycols[res$pvalue>0.05] <- "grey"
```

```
ggplot(res)+
  aes(log2FoldChange,
      -log(padj))+
  geom_point(col=mycols)+
  theme_bw()+
  geom_vline(xintercept=c(-2,2), col="red")+
  geom_hline(yintercept = -log(0.05),col = "red")
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (`geom\_point()`).



Add Annotation

Add gene symbols and entrez ids

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
res$symbol <- mapIds(x=org.Hs.eg.db,
                    keys = row.names(res),
                    keytype = "ENSEMBL",
                    column = "SYMBOL")
```

'select()' returned 1:many mapping between keys and columns

```
res$genename <- mapIds(x=org.Hs.eg.db,
                      keys = row.names(res),
                      keytype = "ENSEMBL",
                      column = "GENENAME")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(x=org.Hs.eg.db,
                    keys = row.names(res),
                    keytype = "ENSEMBL",
                    column = "ENTREZID")
```

'select()' returned 1:many mapping between keys and columns

Pathway Analysis

KEGG Pathway

Run gagw analysis

```
library(gage)
library(gageData)
library(pathview)
```

We need a named vector for fold-change gage

```
data(kegg.sets.hs)
data(sigmet.idx.hs)
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

<NA>	148398	26155	339451	84069	84808
0.17925708	0.42645712	-0.69272046	0.72975561	0.04057653	0.54281049

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
```

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.246882e-03	-3.059466	1.246882e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.066915974	144	1.246882e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

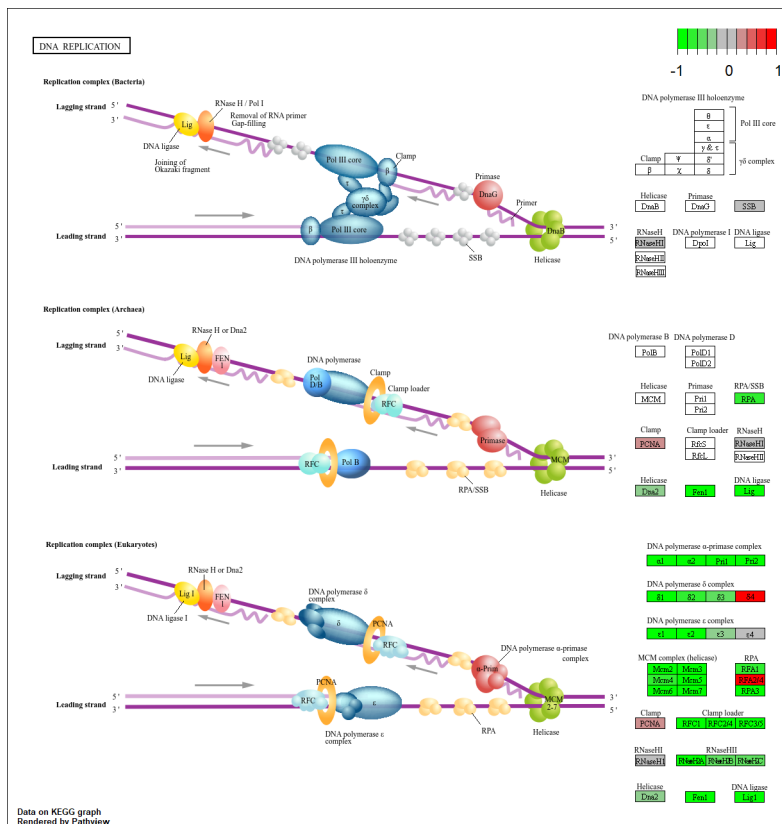
Pathway view

```
pathview(pathway.id = "hsa04110", gene.data = foldchanges)
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/xiaos/OneDrive/ /BIMM143/Class14

Info: Writing image file hsa04110.pathview.png



GO term

Same analysis but using GO genesets rather than KEGG

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets, same.dir=TRUE)

lapply(gobpres, head)
```

\$greater

	p.geomean	stat.mean	p.val
GO:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05

G0:0002009	morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729	tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610	behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562	epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295	tube development	5.953254e-04	3.253665	5.953254e-04

		q.val	set.size	exp1
G0:0007156	homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009	morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729	tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610	behavior	0.1967577	426	1.925222e-04
G0:0060562	epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295	tube development	0.3565320	391	5.953254e-04

\$less

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059	chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236	mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10

		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059	chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236	mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.824205	3.824205
G0:0002009	morphogenesis of an epithelium	3.653886	3.653886
G0:0048729	tissue morphogenesis	3.643242	3.643242
G0:0007610	behavior	3.565432	3.565432
G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

```
head(gobpres$less,4)
```

		p.geomean	stat.mean	p.val
G0:0048285	organelle fission	1.536227e-15	-8.063910	1.536227e-15

G0:0000280	nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067	mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
		q.val	set.size	exp1
G0:0048285	organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280	nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067	mitosis	5.841698e-12	352	4.286961e-15
G0:0000087	M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14

Reactome

Lots of folks like the reactome web interface. You can also run this as an R function but let's look at the web first <https://reactome.org/>

The website wants a text file with one gene symbol per line of the genes you want to map to pathways

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), ]$symbol
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Save our Results

```
write.csv(res, file="myresults.csv")
```